PROBABILITY vs STATISTICS — CONCEPTUAL MCQs

1. You know that a fair die has six equally likely outcomes. You want to calculate the probability of getting a 6. This is an example of:

A. Statistics
B. Probability
C. Regression
D. Sampling

2. You roll a die 1,000 times and record the results. You notice that the number 6 appeared much more frequently than expected. You want to determine whether the die may be biased. This is primarily:

A. Probability
B. Statistics
C. Linear Algebra
D. Calculus

3. Which statement best describes Probability?

A. Using observed data to estimate unknown characteristics
B. Using a known model or assumptions to reason about possible outcomes
C. Organizing data into tables
D. Calculating averages from collected data

4. Which statement best describes Statistics?

A. Predicting possible outcomes from a known probability model
B. Using observed data to understand or estimate an underlying process
C. Generating random numbers
D. Calculating only probabilities between 0 and 1

5. A company knows that historically 2% of its products are defective. It wants to estimate how many defective products might appear in the next 10,000 products.

This is primarily:

A. Statistics
B. Probability
C. Data cleaning
D. Visualization

6. A company collects data from 10,000 products and finds that 250 are defective. It wants to estimate the defect rate of the production process.

This is primarily:

A. Probability
B. Statistics
C. Linear Algebra
D. Optimization

7. Which direction best represents Probability?

A. Data → Model
B. Model → Possible outcomes
C. Data → Visualization
D. Sample → Database

8. Which direction best represents Statistics?

A. Model → Possible outcomes
B. Possible outcomes → Model
C. Observed data → Understanding/inference about the underlying process
D. Probability → Randomness

9. A medical researcher knows that a disease occurs in approximately 1% of the population. She wants to calculate the probability that a randomly selected person has the disease.

This is primarily:

A. Statistics
B. Probability
C. Data preprocessing
D. Regression

10. A medical researcher observes that 15 out of 1,000 randomly selected people have a particular disease. She wants to estimate the disease prevalence in the larger population.

This is primarily:

A. Probability
B. Statistics
C. Matrix algebra
D. Optimization

11. Consider the following situation:

"A coin is believed to be fair. What is the probability of getting 7 Heads when the coin is tossed 10 times?"

This is:

A. Statistics
B. Probability
C. Descriptive statistics
D. Sampling

12. Now consider:

"A coin is tossed 10 times and produces 9 Heads. Based on this observation, determine whether the coin is likely to be fair."

This is:

A. Probability
B. Statistics
C. Matrix multiplication
D. Data visualization

13. Which question is a Probability question?

A. What is the average height of students in this class?
B. What is the probability that a randomly selected student is taller than 180 cm?
C. What is the median height of the students?
D. What is the standard deviation of student heights?

14. Which question is a Statistics question?

A. What is the probability of getting Heads on a fair coin?
B. What is the probability of rolling a 6?
C. What is the average income observed in this dataset?
D. What is the probability of rain tomorrow?

15. A weather model says that there is a 70% probability of rain tomorrow.

This statement is an example of:

A. Statistics
B. Probability
C. Descriptive statistics
D. Sampling

16. You collect the rainfall measurements from the last 20 years and calculate the average annual rainfall.

This activity is primarily:

A. Probability
B. Statistics
C. Probability distribution
D. Random sampling

Just to make it clearer – Probability and statistics

17. "What is likely to happen?"

A. Probability
B. Statistics

18. "What happened in the data?"

A. Probability
B. Statistics

19. "What can we learn about the population from our sample?"

A. Probability
B. Statistics

20. "If the assumptions are known, what outcomes should we expect?"

A. Probability
B. Statistics